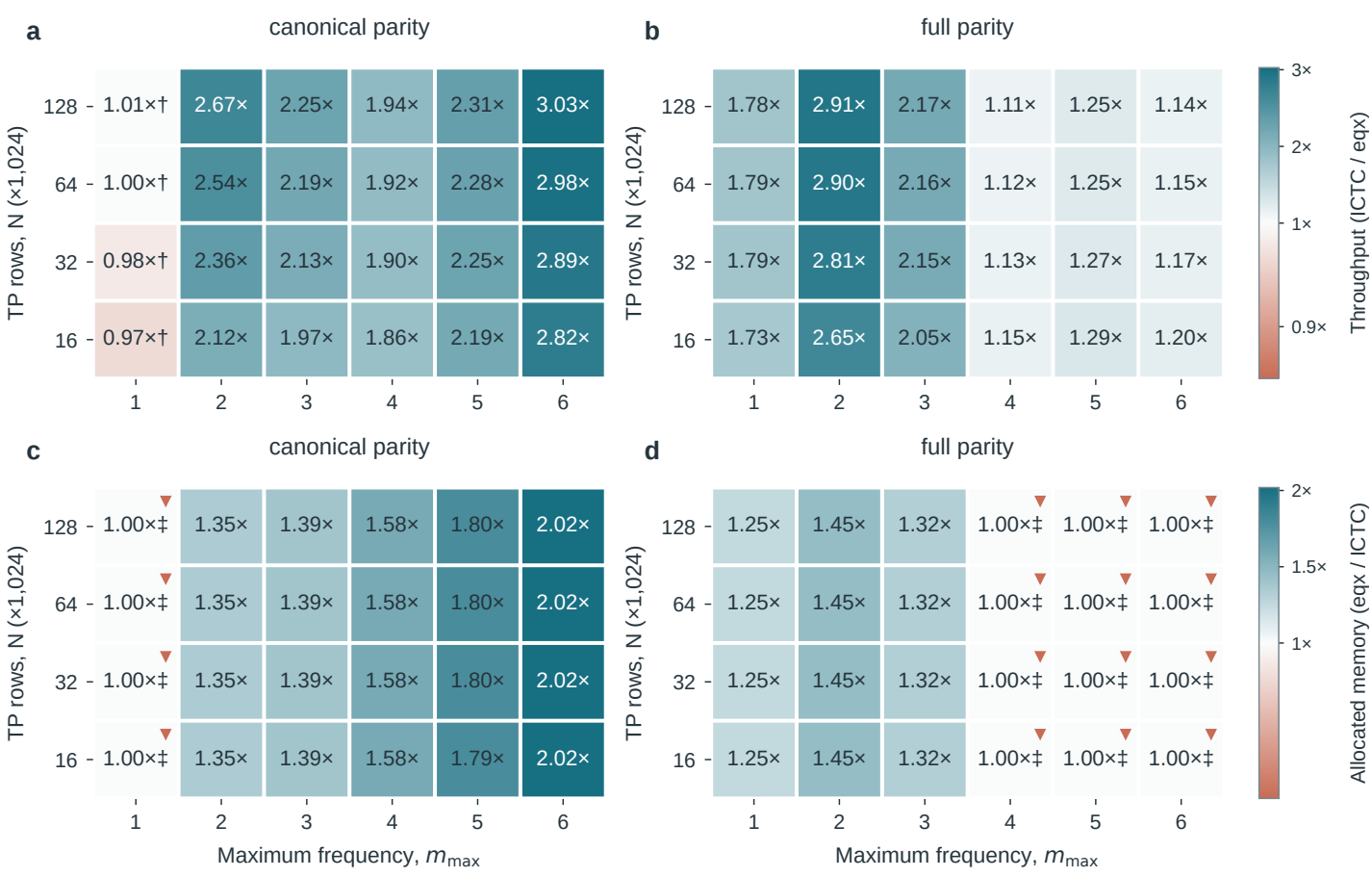

ICTC-O2 HP vs eqx: throughput and peak memory
O(2) tensor products · Forward · C = 96 · N = 16,384–131,072 · m_max = 1–6
a
canonical parity
b
full parity
c
canonical parity
d
full parity
TP rows, N (×1,024)
128
64
32
16
1.01×† 2.67× 2.25× 1.94× 2.31× 3.03×
1.00×† 2.54× 2.19× 1.92× 2.28× 2.98×
0.98×† 2.36× 2.13× 1.90× 2.25× 2.89×
0.97×† 2.12× 1.97× 1.86× 2.19× 2.82×
1.78× 2.91× 2.17× 1.11× 1.25× 1.14×
1.79× 2.90× 2.16× 1.12× 1.25× 1.15×
1.79× 2.81× 2.15× 1.13× 1.27× 1.17×
1.73× 2.65× 2.05× 1.15× 1.29× 1.20×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.79× 2.02×
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1 2 3 4 5 6
Maximum frequency, m_max
Throughput (ICTC / eqx)
3×
2×
1×
0.9×
Allocated memory (eqx / ICTC)
2×
1.5×
1×
† IQR overlap * isolated timing ‡ memory increase
v0.1.5